endosperm of coconut) which contains cell division inducing substances. In 1954, Skoog and Miller discovered that autoclaved DNA from herring sperm stimulated cell division in tobacco pith cells. They called this cell division inducing principle as kinetin (chemical structure: 6-Furfuryl Amino Acid). This does not occur in plants. In 1963, Letham introduced the term cytokinin. In 1964, Letham and Miller isolated and identified a new cytokinin called **Zeatin** from unripe grains of maize. The most widely occurring cytokinin in plants is Iso Pentenyl adenine (IPA).

2. Occurrence

Cytokinin is formed in root apex, shoot apex, buds and young fruits.

3. Precursor

Cytokinins are derivatives of the purine adenine.

4. Bioassay (Neem Cotyledon Assay)

Neem cotyledons are measured and placed in cytokinin solution as well as in ordinary water. Enlargement of cotyledons is an indication of cytokinin activity.

5. Transport in plants

The distribution of cytokinin in plants is not as wide as those of auxin and gibberellins but found mostly in roots. Cytokinins appear to be translocated through xylem.

6. Physiological effect

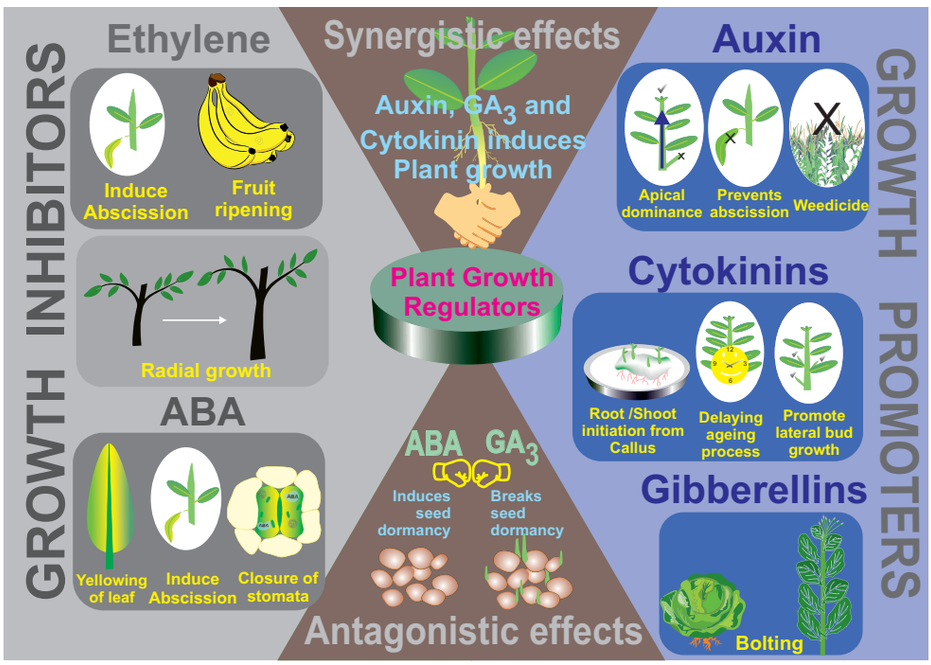
- Cytokinin promotes cell division in the presence of auxin (IAA).
- Cytokinin can break the dormancy of certain light-sensitive seeds like tobacco and induces seed germination.
- Cytokinin promotes the growth of lateral bud in the presence of apical bud.
- Application of cytokinin delays the process of aging by nutrient mobilization. It is known as Richmond Lang effect.
- Cytokinin (i) increases rate protein synthesis (ii) induces the formation of inter-fascicular cambium (iii) overcomes apical dominance (iv) induces formation of new leaves, chloroplast and lateral shoots.
- Plants accumulate solutes very actively with the help of cytokinins.

15.2.4 Ethylene (Gaseous Phytohormone)

Almost all plant tissues produce ethylene gas in minute quantities.

1. Discovery

In 1924, **Denny** found that ethylene stimulates the ripening of lemons. In 1934, **R. Gane** found that ripe bananas contain abundant ethylene. In 1935, **Cocken** *et al.,* identified ethylene as a natural plant hormone.



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